

Performance test coverage — lab replication report

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Repository and scope

The workspace was already a clean clone of <https://github.com/SpencerLabAQ/performance-test-coverage> at commit 927e8dc4a869868669de00cb971549006cd2446b. This existing clone was used. All four top-level analysis scripts were executed from analysis/. The nested plot/table scripts were also tested separately. These are analyses of supplied CSV data, rather than fresh Java benchmark or JUnit executions.

Environment and initial failures

README requires Python 3.10.5 with pinned 2023 scientific libraries. Only Python 3.14.7 was installed. Each of the four initial commands failed with ModuleNotFoundError: No module named pandas. A project-local .venv was created and compatible current libraries installed. Windows application control blocked a pandas DLL inside the sandbox; execution outside the sandbox was attempted with approval. The environment is therefore a compatibility replication, not an exact reproduction of the documented environment.

Exact installed versions are preserved in environment.txt. Initial failure evidence is in initial-environment/. The analysis source files were left unchanged.

Conclusion after each main test

1-Overlap-JMH.py

Command: cd analysis; python 1-Overlap-JMH.py. Exit status: 0. Elapsed: 11.713 seconds.

Working: completed successfully and generated the expected CSV.

Evidence: 1-Overlap-JMH.py.log

2-Overlap-JUnit.py

Command: cd analysis; python 2-Overlap-JUnit.py. Exit status: timeout. Elapsed: 300.035 seconds.

Did not complete within the 300-second test limit. This is a runtime limitation in this run, not proof of an incorrect formula. The source enumerates all pairs of tests and repeatedly indexes pandas Series, making large suites expensive. The selected values were independently verified below. The JUnit CSV remained the pre-existing reference; it was not freshly generated by this command.

Evidence: 2-Overlap-JUnit.py.log

3-Coverage-JMH.py

Command: cd analysis; python 3-Coverage-JMH.py. Exit status: 0. Elapsed: 5.449 seconds.

Working: completed successfully and generated the expected CSV. No terminal table is expected: the script contains a bare dataframe expression and writes the results to CSV.

Evidence: 3-Coverage-JMH.py.log

4-Coverage-JUnit.py

Command: cd analysis; python 4-Coverage-JUnit.py. Exit status: 0. Elapsed: 6.388 seconds.

Working: completed successfully and generated the expected CSV. No terminal table is expected: the script contains a bare dataframe expression and writes the results to CSV.

Dependency limitation: this successful run reads the existing Overlap_Ratios_Junit.csv after the overlap command timed out. Coverage counts and percentages were freshly transformed, but its overlap column comes from the supplied reference CSV.

Evidence: 4-Coverage-JUnit.py.log

Fresh outputs versus repository reference outputs

Output	Rows	Changed cells	Missing projects
Overlap_Ratios_JMH.csv	28	0	None
Overlap_Ratios_Junit.csv	19	0	None
coverageJMH.csv	28	0	None
coverageJUnit.csv	19	0	None

A matching pre-existing CSV alone does not prove successful execution. Interpret this table together with each command's exit status. The baseline files are replication-package references, not independently measured paper values.

Selected overlap metrics and paper comparison

Project	JMH overlap	JUnit overlap	Paper comparison
byte-buddy	1.00	0.70	JMH 1.00 explicitly stated (§4.1); JUnit shown in Fig. 8.
fastjson	0.91	0.15	JMH 0.91 and JUnit 0.15 explicitly stated (§4.1).
commons-bcel	0.09	0.66	JMH 0.09 and JUnit 0.66 explicitly stated (§4.1).
SquidLib	0.80	0.50	Both shown graphically in Fig. 8; exact decimals not stated in prose.

The five values stated explicitly in the paper are compared directly below; other exact values above are compared to the saved repository baseline. No exact decimals are inferred from chart heights.

Project	Suite	Paper	Output	Conclusion
byte-buddy	JMH	1.00	1.00	Match
fastjson	JMH	0.91	0.91	Match
fastjson	JUnit	0.15	0.15	Match
commons-bcel	JMH	0.09	0.09	Match
commons-bcel	JUnit	0.66	0.66	Match

Source: <https://link.springer.com/article/10.1007/s10664-025-10712-3> (published 30 August 2025), §3.2.1, §4.1, Figures 5–8. The final published version is used because its title matches the repository README. If the lab uses another version, compare that version separately.

Independent validation of overlap calculations

Project	Suite	Tests	Shared methods	Union	Raw ratio	CSV	Check
byte-buddy	JMH	39	107	107	1.000000	1.00	Match
fastjson	JMH	4	58	64	0.906250	0.91	Match
commons-bcel	JMH	3	36	396	0.090909	0.09	Match
SquidLib	JMH	236	353	441	0.800454	0.80	Match
byte-buddy	JUnit	6357	454	646	0.702786	0.70	Match
fastjson	JUnit	4979	887	6053	0.146539	0.15	Match
commons-bcel	JUnit	137	712	1087	0.655014	0.66	Match
SquidLib	JUnit	73	175	351	0.498575	0.50	Match

Independent check: construct a set of covered method identifiers for each distinct test, count each identifier across tests, and divide identifiers present in at least two tests by all identifiers covered. This implements the paper’s definition without enumerating all pairs. The original pairwise algorithm is quadratic in test count. For a suite with fewer than two tests it returns zero; missing Methods values may need additional filtering for other datasets.

Coverage tests: selected projects

Project	Suite	Total methods	Covered	Directly covered	Coverage %	Direct %	Scope
byte-buddy	JMH	5046	107	23	2.12	0.46	37.95
byte-buddy	JUnit	5046	646	486	12.8	9.63	3.05
fastjson	JMH	17524	64	3	0.37	0.02	30.5
fastjson	JUnit	17524	6053	4879	34.54	27.84	13.12
commons-bcel	JMH	3132	396	106	12.64	3.38	144.67
commons-bcel	JUnit	3132	1087	133	34.71	4.25	33.44
SquidLib	JMH	5663	441	194	7.79	3.43	44.53
SquidLib	JUnit	5663	351	173	6.2	3.05	11.81

Coverage percentages use the supplied summary counts: $100 \times \text{covered methods} / \text{total methods}$. Direct coverage uses directly covered methods. Scope is the average methods per test. Values are rounded to two decimals. These test the CSV transformation and summaries, not the correctness or completeness of the original profiling.

Independent arithmetic checks passed for both coverage percentages and scope in all eight selected project/suite rows. SquidLib has JMH coverage 7.79% versus JUnit 6.20%, and direct coverage 3.43% versus 3.05%, agreeing with the paper's qualitative exception.

Plot and table script tests

rq1-fig3.py: exit 0, 82.049 s. Working: completed with the noninteractive Agg backend.

rq1-fig4.py: exit 0, 4.502 s. Working: completed with the noninteractive Agg backend.

rq1-fig5.py: exit 0, 16.269 s. Working: completed with the noninteractive Agg backend.

rq1-table_II.py: exit 1, 1.393 s. Not working: TypeError: Invalid value '-' for dtype 'float64'

rq2-fig6.py: exit 0, 6.095 s. Working: completed with the noninteractive Agg backend.

rq2-fig7.py: exit 0, 6.068 s. Working: completed with the noninteractive Agg backend.

rq2-fig8.py: exit 0, 6.065 s. Working: completed with the noninteractive Agg backend.

rq3-fig10.py: exit 0, 17.085 s. Working: completed with the noninteractive Agg backend.

rq3-fig9.py: exit 0, 7.705 s. Working: completed with the noninteractive Agg backend.

rq4-fig11.py: exit 0, 7.578 s. Working: completed with the noninteractive Agg backend.

rq4-fig12.py: exit 0, 7.364 s. Working: completed with the noninteractive Agg backend.

rq4-figxx.py: exit 0, 6.949 s. Working: completed with the noninteractive Agg backend.

rq4-figxy.py: exit 0, 6.941 s. Working: completed with the noninteractive Agg backend.

The README names rq2-fig9.py, rq3-fig11.py and rq4-fig13.py, which are absent. Actual filenames were discovered from disk. Plot numbering in this package differs from final-paper numbering; compare by metric/title.

Table-script diagnosis: `df_final.fillna("-")` attempts to insert text in float columns, which pandas 3 rejects. Use the README's original Python/pandas environment, or explicitly convert display columns to object before filling them in a future compatibility fix. The script also invokes `pdflatex`, which was not found on this machine; this is an additional requirement for completing PDF generation after the pandas issue is resolved.

Overall conclusion for the lab

Three of four main commands completed successfully. The JUnit overlap command exceeded the chosen five-minute limit. All eight selected overlap ratios independently matched the supplied results; five also matched exact values explicitly stated in the paper. Twelve of thirteen nested plot/table scripts completed; the table script failed under pandas 3. The remaining work for exact-environment replication is to run the original Python 3.10.5/pinned libraries, provide pdflatex, and allow enough time for the full JUnit pairwise calculation.

Use the command outcomes above to determine execution success. The selected overlap values agree with the paper's five explicit numerical claims when the Match checks pass. High overlap means repeated method coverage, not necessarily unnecessary tests: different workloads may intentionally exercise the same methods. The selected coverage results can also be checked against the paper's qualitative finding that SquidLib has greater JMH than JUnit coverage. Exact agreement with bundled outputs verifies reproduction from the supplied data; it does not constitute independent collection of Java call stacks or verification of the paper's machine-learning experiments.

How to repeat

From the repository root: `.\.venv\Scripts\python.exe lab-report\run_analysis.py`

Then: `.\.venv\Scripts\python.exe lab-report\run_plots.py`

Then: `.\.venv\Scripts\python.exe lab-report\build_report.py`

The runner executes the four main scripts sequentially from analysis/ and preserves logs. Output baselines are preserved only on the first run.